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Research Project on the Topic:

**Investigation of the Stability of Interpreted Credit Scoring Models Under Various
Coding and Feature Selection Strategies**

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Abstract

The project consistently performs data loading and analysis, processing of missing and abnormal values, generation of derived features, separation of features into numerical and categorical ones, as well as the construction of a preprocessing pipeline. Special attention is paid to comparing approaches to encoding categorical variables, including One-Hot Encoding, Ordinal Encoding, Target Encoding, and CatBoost Encoding.

To solve the problem, several models were trained and compared: logistic regression, decision tree, LightGBM, MLP Small, and MLP Deep. The quality of the models was assessed using ROC-AUC, Average Precision, LogLoss, and Accuracy metrics. Additionally, informative features were selected using SelectKBest, permutation importance, and SHAP.

The practical value of the work is related to the fact that the resulting pipeline allows not only to predict the probability of default, but also to explain the result at the level of individual features. The interpretation is performed using logistic regression coefficients, decision tree rules, SHAP values, feature importance, and partial dependence plots. An assessment of the stability of the main models and feature importance for different data splits was also carried out. An additional part of the work is the construction of an LLM-assisted explanation layer. The language model is not used as a separate scoring model, but as a tool for transforming model outputs and SHAP-based local explanations into a clear textual summary. This makes it possible to present the result of the scoring model in a more understandable form for a human user while preserving the connection with the original model output.

As a result, it is shown that the interpretable approach to credit scoring can provide acceptable forecast quality and at the same time maintain the transparency necessary for practical application in the financial sector. LightGBM demonstrated the best predictive quality, logistic regression remained a strong interpretable baseline, MLP models acted as additional nonlinear benchmarks, and the LLM-assisted layer provided an additional communication level for explaining scoring results.

Keywords

Credit scoring, credit risk, default, interpretable machine learning, binary classification, logistic regression, decision tree, LightGBM, feature selection, categorical feature encoding, SHAP, model stability, LLM, MLP Small, MLP Deep.

1 Introduction

The topic of the thesis is related to the construction of an interpretable credit scoring model. The choice of this task is due to the fact that this dataset reflects the real features of credit analysis well, namely the presence of numerical and categorical features, missing values, abnormal values, class imbalance, and complex dependencies between features and the target variable. Working with such material makes it possible to consider not a separate technical technique, but a holistic project aimed at creating a scoring pipeline, from data loading to interpretation and verification of result stability.

1.1 Relevance

In recent years, financial institutions have increasingly used machine learning methods to assess their clients' creditworthiness. This is due to the growing volume of data, the need to make decisions faster, and the desire to reduce credit risks. Automated scoring systems make it possible to process a large number of applications, take into account dozens of factors simultaneously, and find dependencies that are difficult to identify through manual analysis. However, as the accuracy of models increases, another problem arises. The more complex the algorithm, the more difficult it is to understand why a positive conclusion was made on one application and a negative one on another. This situation is particularly sensitive in the banking sector, since lending decisions affect both the interests of the organization and the interests of the client. In addition, under strict regulation, it is not enough to obtain a good forecast on its own. It is equally important to explain the logic of the model, show which features influenced the result, and assess how stable the model remains when input data and data processing methods change.

That is why interpretable approaches to credit scoring are of particular interest. Logistic regression, decision trees, and other transparent models make it possible not only to estimate the probability of default, but also to analyze the decision-making mechanism at the level of individual features [6]. However, even for such models, the issue of reliability remains crucial. If the result changes markedly with a different categorical feature encoding strategy, with a new sample split, or with a small change in the set of factors, then the practical value of the model decreases. As a result, the task of building not just an interpretable, but also a stable scoring model remains relevant.

1.2 Aim and Objectives

The aim of the thesis is to develop an interpretable credit scoring pipeline based on data and to evaluate its quality, transparency, and stability using different approaches to feature processing and model construction.

To achieve this goal, the following tasks were formulated:

1. Analyze the subject area of credit scoring and the role of interpretable models in financial decisions;
2. Consider the features of provided dataset;
3. Perform a preliminary data analysis, process missing and abnormal values, and generate derived features;
4. Divide the features into numerical and categorical ones and build a preprocessing pipeline;
5. Compare several encoding strategies for categorical variables;
6. Train and compare multiple classification models, including logistic regression, decision tree, LightGBM, MLP Small, and MLP Deep;
7. Select informative features;
8. Interpret the results using model coefficients, decision tree rules, permutation importance, SHAP, and partial dependence plots;
9. Evaluate the stability of model quality, predictions, and feature importance across different data splits;
10. Develop an LLM-assisted explanation layer for transforming model outputs and local interpretation results into a clear textual explanation.

1.3 Object and Subject

The object of the thesis is the process of assessing borrowers' credit risk using machine learning methods.

The subject of the thesis is methods for constructing, interpreting, and verifying the stability of credit scoring models based on tabular data.

1.4 Methods Used

The paper uses methods of preliminary data analysis, statistical feature processing, construction and comparison of binary classification models, cross-validation, feature selection, and interpretation of machine learning results. The preprocessing stage included processing missing and abnormal values, generating derived financial indicators, separating numerical and categorical features, scaling numerical variables, and encoding categorical variables using One-Hot Encoding, Ordinal Encoding, Target Encoding, and CatBoost Encoding.

Several classification models were used to solve the credit scoring task: logistic regression, decision tree, LightGBM, MLP Small, and MLP Deep. Logistic regression and decision tree were considered as interpretable baseline models, LightGBM was used as a strong gradient boosting model for tabular data, and MLP models were added as nonlinear neural network benchmarks. The metrics ROC-AUC, Average Precision, LogLoss, and Accuracy were used to evaluate the quality of the models. Feature selection and interpretation were carried out using SelectKBest, permutation importance, SHAP values, model coefficients, decision tree rules, and partial dependence plots. Logistic regression coefficients were used to analyze the direction and strength of the influence of features, decision tree rules were used for rule-based interpretation, and SHAP values were used to explain the contribution of individual features to model predictions.

The stability check was carried out taking into account different sample splits and repeated runs of the main models. The analysis included the stability of quality metrics, feature importance, and model predictions. In addition, an LLM-assisted explanation layer was implemented. The language model was used not for predicting default probability, but for generating textual explanations based on the predicted probability, risk category, and SHAP-based local feature contributions.

1.5 Novelty

The novelty of the work lies in the fact that the emphasis is placed not only on comparing the quality of the models, but also on comparing the interpretability and sustainability of the results. The project consistently combines the stages of data preprocessing, comparing methods for encoding categorical features, selecting factors, interpreting models, and verifying the stability of their behavior. This allows us to consider the final model not as a separate algorithm, but as an integrated scoring decision-making system.

2 Theoretical foundations of Credit scoring and Interpreted machine learning

2.1 Credit scoring as a credit risk assessment tool

Credit scoring occupies an important place in the modern borrower assessment system. This concept usually refers to an approach in which the probability of a client fulfilling or not fulfilling his obligations is determined based on a set of quantitative and qualitative characteristics. The result of such an assessment is a scoring score or the probability of default, which is then used to make a decision on granting a loan, changing its terms, or rejecting the application [7].

The need to use scoring is related to the very nature of credit risk. Any credit institution, when issuing borrowed funds, is faced with uncertainty: it is impossible to determine in advance with full confidence whether the client will fulfill the obligations on time and in full. The risk of non-repayment is formed under the influence of many factors, such as income level, employment, debt burden, credit history, family composition, age, type of employment, behavior on previous obligations, and other characteristics. If such decisions are made only manually, the process becomes slow, expensive, and largely dependent on the subjective assessment of a specialist. The scoring approach allows you to translate the loan decision into a more formalized form. To do this, the customer data is compared with patterns identified by historical examples. If certain combinations of features in the archived data were more often associated with delay or default, the model captures this relationship and uses it when evaluating new applications. Due to this, it is possible not only to speed up the processing of applications, but also to achieve a more uniform approach to customer assessment.

In practice, credit scoring solves several tasks at once. Firstly, it helps to separate borrowers by the level of risk. Secondly, it makes possible the mass processing of applications without the constant participation of an analyst at each stage. Thirdly, the scoring system is used not only at the time of loan issuance, but also in the future, for example, when monitoring a portfolio, reviewing limits, identifying customers with an increased probability of delinquency and forming an organization's risk strategy. In other words, scoring has long ceased to be just a primary selection tool and has become part of a broader credit risk management system.

Classical credit scoring is based on tabular data about the borrower. Socio-demographic characteristics, information about employment, financial status, credit activity, and previous behavior of the client are most often used. Binary classification methods are particularly well suited for working with such data, in which the target variable reflects the presence or absence of a

default. Within the framework of this formulation, the task is to determine by a set of signs the probability that the borrower will fall into the class of problem clients.

One of the key advantages of scoring is the ability to quantify risk. In this case, the decision is made not on the level of the overall impression of the client, but on the basis of a predefined model. This reduces the influence of random factors and makes the procedure more reproducible. In addition, scoring models allow you to compare different segments of customers with each other, set risk thresholds and select a credit policy based on the acceptable level of losses. At the same time, a high scoring score by itself does not mean absolute reliability of the client, but a low score indicates a guaranteed default. It's always about probabilistic estimation. The model does not predict the future in the strict sense, but only determines how much the behavior of a particular borrower is similar to the behavior of those customers who have previously experienced repayment problems. For this reason, the quality of scoring directly depends on the data set, the correctness of the preparation of features, the choice of the model, and how well the current situation corresponds to the conditions under which the model was trained.

Historically, simple and well-explained models have been widely used in banking practice, primarily logistic regression and scoring maps [6]. Their popularity was associated not only with the acceptable quality of the forecast, but also with the transparency of the results. It is important for a specialist to understand why one client received a high risk, while the other, on the contrary, received a low one. In the context of regulation, internal audit, and the need to justify credit decisions, such transparency is particularly valuable. For this reason, even with the proliferation of more complex algorithms, interpreted models remain of practical importance. At the same time, modern credit scoring tasks are becoming increasingly complex. The amount of data is increasing, the feature structure is becoming more complex, and the requirements for forecast quality are increasing. This led to an expansion of the set of methods used: in addition to classical statistical models, decision trees, ensemble algorithms, and gradient boosting began to be actively used. They often provide higher accuracy, but at the same time reduce the transparency of the assessment. As a result, there is a contradiction between the quality of the forecast and the explainability of the model in scoring projects.

The sustainability of scoring solutions remains a separate problem. Even an interpreted model can behave erratically if its results strongly depend on the way categorical features are encoded, on a specific sample split, or on the composition of the training data. This is especially important for the credit sector, since the system must not only show good metrics on a test sample, but also demonstrate reliable behavior in repeated launches and on new data. Therefore, credit scoring should be considered not just as a classification model, but as a risk management tool that

requires quality, transparency, and reproducibility.

As a result, credit scoring is one of the basic tools for assessing credit risk, allowing to formalize work with borrowers, speed up decision-making and reduce the proportion of subjective assessments. Its practical value is determined not only by the accuracy of the forecast, but also by the extent to which the model is understandable, stable and suitable for use in real banking conditions.

2.2 Data features in credit scoring tasks

The data in the credit scoring tasks have several important features that directly affect the construction of the model and the quality of the forecast. Most often, work is done with tabular data, where each row corresponds to a separate client, and the attributes describe his socio-demographic, financial, and behavioral characteristics.

One of the main features of such data is the combination of numerical and categorical features. Numeric values include, for example, age, income, loan amount, and annuity amount. Categorical features reflect the type of employment, marital status, education level, housing conditions, and other characteristics of the client. This requires separate processing of different groups of features and choosing the appropriate way to encode categorical variables.

Scoring data is also characterized by omissions and abnormal values. Omissions may occur due to incomplete questionnaires, errors in filling out, or information storage features. Abnormal values are often associated with either rare cases or technical errors. If such data is not processed in advance, it can significantly distort the simulation result.

Another important feature is the class imbalance. In practice, there are usually significantly fewer customers in default than there are customers who meet their obligations without delay. Because of this, estimating the model based on accuracy alone turns out to be insufficient, since high formal accuracy does not always mean good identification of risky borrowers. The need to generate new features also plays an essential role. In many cases, it is not the original variable itself that is more useful, but a derived indicator, such as the ratio of income to the loan amount or the share of an annuity in income. This allows you to accurately reflect the financial situation of the client and improve the quality of the forecast. In addition, scoring data is sensitive to the preprocessing method. The processing of omissions, scaling of numerical features, encoding of categorical variables, and selection of factors can significantly affect both the final metrics and the interpretability of the model.

As a result, the data in credit scoring tasks is a heterogeneous and rather complex mate-

rial. Their specificity is related to the combination of different types of features, the presence of omissions and anomalies, the imbalance of classes and the need to construct new factors. That is why the quality of the scoring model is largely determined not only by the chosen algorithm, but also by how well the data preparation is performed.

2.3 Machine learning methods for scoring models

In credit scoring tasks, machine learning methods are used to estimate the probability of default based on customer attributes. Most often, such a task is considered as a binary classification, where the model should assign the borrower to a low or high risk group.

Logistic regression remains one of the most common methods [2]. It is widely used in scoring tasks due to its simplicity, stability and high interpretability. The result of such a model can be explained through feature coefficients, which is especially important in the banking sector, where transparency in decision-making is required. At the same time, logistic regression reflects complex nonlinear relationships between features worse.

Decision trees are another important method. They make a forecast based on consistent conditions based on the characteristics of the client. The advantage of the tree is clarity: it is possible to trace which rules lead to the final decision. However, such models are often sensitive to changes in data and can be retrained without additional configuration. Ensemble methods, in particular gradient boosting, are also used to improve the quality of forecasting in scoring tasks. One of the most popular implementations is LightGBM. Such models usually show higher accuracy on tabular data, as they are able to account for complex dependencies and interactions between features. At the same time, their interpretation turns out to be noticeably more difficult compared to logistic regression and decision trees.

Other approaches can be used in a number of tasks, such as random forest, GAM, or EBM. Their application depends on the data structure, forecast quality requirements, and the level of explainability required. When choosing a method for a scoring model, it is important to take into account not only the final metrics, but also the practical requirements for the model. In credit scoring, accuracy, interpretability, stability, and usability in real-world conditions are important. Therefore, in such projects, several algorithms are usually compared at once, after which an approach is chosen that provides the most successful balance between the quality of the forecast and the transparency of the result.

2.4 Approaches to coding categorical features

Categorical features also occupy an important place, since many characteristics of a client are determined not by a number, but by belonging to a certain group. These criteria include gender, marital status, type of employment, level of education, type of housing, and other parameters. However, most machine learning algorithms cannot work with such data directly, so categorical variables must be converted to numerical form before training the model.

One of the most common methods is One-Hot Encoding. With this approach, a separate binary attribute is created for each category, which takes the value 0 or 1. This method is well suited for features with a small number of unique values and does not set an artificial order between categories. Its disadvantage is that with high cardinality, the number of features increases rapidly, which can make the model more cumbersome. Another option is Ordinary Encoding, in which each category is assigned a numeric code. This method is easy to implement and does not increase the size of the data. However, it has a limitation: the model can perceive assigned codes as ordered values, although in reality there may not be such an order between categories. For this reason, this approach requires careful application. Scoring tasks often use more flexible methods, such as Target Encoding. In this case, the category is replaced by statistics related to the target variable, for example, the average default rate in this group. This approach allows us to take into account the relationship between category and risk, but if applied incorrectly, it can lead to overfitting, especially in small samples. A similar method is CatBoost Encoding, which also uses information about the target variable, but it does it more steadily and accurately, reducing the risk of data leakage. This method often shows good results on tabular data with a large number of categorical features.

The choice of coding method affects not only the quality of the forecast, but also the interpretability of the model. For some algorithms, One-Hot Encoding is more convenient, while for others, Target Encoding or CatBoost Encoding is better. As a result, scoring projects usually consider several approaches, after which the option that provides the most successful balance between accuracy, stability and convenience of further analysis is selected.

2.5 Methods of selection and interpretation of features

It is very important to understand which signs really affect the result. A large number of variables does not always improve the quality of the forecast. On the contrary, redundant or poorly informative features can complicate the model, reduce its stability, and make interpretation more difficult. For this reason, scoring projects usually use a selection of the most significant factors.

One common approach is SelectKBest. This method allows you to select the features that are most strongly associated with the target variable according to a given statistical criterion. Its advantage lies in the simplicity and convenience of the initial selection, especially at the stage of preliminary data preparation.

Permutation Importance is considered another important method [4]. In this case, the significance of a feature is assessed by how much the quality of the model deteriorates after random mixing of its values. If the metric decreases noticeably after such a step, the feature can be considered important. This approach is convenient because it allows you to evaluate the contribution of features after training the model.

SHAP values are often used for deeper interpretation [5]. They show how much each feature contributes to the final forecast for a particular object. Using this method, it is possible to analyze both the overall importance of factors and the influence of features on individual predictions. In credit scoring tasks, this is especially useful because it allows you to explain why the model estimated the risk of a particular client in this way.

Additionally, Partial Dependence plots are used for interpretation. They allow you to see how the model's forecast changes when one feature changes, all other things being equal. This approach helps to understand not only the significance of the feature, but also the nature of its impact on the probability of default.

Embedded analysis methods are also used for interpreted models. In logistic regression, coefficients of features are estimated, which can be used to judge the direction and strength of the influence of factors. In decision trees, interpretation is based on the rules of partitioning and the structure of the tree. This makes it possible to explain the model without using complex external methods.

2.6 Metrics for evaluating the quality of models

Evaluating the quality of models requires the use of several metrics simultaneously. This is due to the fact that one metric does not always allow us to fully evaluate the behavior of the model, especially if there is an imbalance of classes, when there are significantly fewer customers with default than customers without delay.

One of the main metrics is ROC-AUC. It shows how well the model separates objects of two classes and is able to distinguish more risky borrowers from less risky ones. The higher the ROC-AUC value, the better the model ranks customers according to the probability of default. This metric is widely used in scoring tasks because it allows you to evaluate the quality of a model

regardless of a specific classification threshold.

Another important metric is considered to be Average Precision. It is especially useful in conditions of unbalanced data, when it is necessary to assess how accurately the model identifies customers with a high risk of default. Unlike ROC-AUC, this metric focuses more on the quality of positive class recognition.

LogLoss is used to evaluate probabilistic forecasts. This metric takes into account not only the correctness of the classification, but also the confidence of the model in its predictions. If the model makes an overly confident but erroneous forecast, the LogLoss value increases. Due to this, it is possible to understand how correctly the model estimates the probability of default, and not just the final class.

Additionally, Accuracy is often used, that is, the proportion of correctly classified objects. This metric is simple and clear, but in scoring tasks it is usually not enough to consider it separately. With a pronounced class imbalance, high accuracy can be achieved even if the model does not identify customers with default well.

3 Data preparation and pipeline construction of processing

The implementation of all experimental procedures described in this work is available in the GitHub repository [3].

3.1 Characteristics of the dataset

The anonymized data approximating reality was chosen as the basis for the practical part. It is well suited for credit scoring tasks because it contains information about loan applications and allows you to solve the binary classification problem, where it is necessary to assess the likelihood of payment difficulties for the borrower. The target variable in the training sample is set by the TARGET attribute: a value of 0 corresponds to applications without fixed repayment problems, and a value of 1 is associated with an unfavorable outcome on the loan.

The data set has a relational structure rather than a flat one. It consists of seven interconnected tables, each of which describes the borrower or his credit history from different angles. The basic tables are `application_train` and `application_test`, where information on current applications is collected. Each application is identified by the `SK_ID_CURR` key, and the TARGET variable is additionally present in the training part. In addition to the main table, the set includes data on the client's previous loans with other financial institutions, the history of balances on

these loans, previous applications within the company, monthly records on POS and cash loans, credit card information, as well as the history of payments on previously issued loans. Due to this, the same client can be described not by one line, but by a whole system of related records.

The `application_train` table is central to the work, since it contains the target variable and the main set of features for building a scoring model. In the sample used, this table includes 307,511 observations and 122 features (Figure 3.1). One row corresponds to one loan application. The fields in the table describe the client from different angles: here you can find information about the type of loan agreement, gender, availability of a car and real estate, number of children, income level, loan amount, annuity amount, as well as a number of other socio-demographic and financial characteristics. This structure already shows at the first stage that the work is being carried out with a large and richly rich tabular data set.

Size application_train: (307511, 122)

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY
0	100002	1	Cash loans	M	N	Y	0	202500.0	406597.5	24700.5
1	100003	0	Cash loans	F	N	N	0	270000.0	1293502.5	35698.5
2	100004	0	Revolving loans	M	Y	Y	0	67500.0	135000.0	6750.0
3	100006	0	Cash loans	F	N	Y	0	135000.0	312682.5	29686.5
4	100007	0	Cash loans	M	N	Y	0	121500.0	513000.0	21865.5

5 rows x 122 columns

Figure 3.1: A fragment of the `application_train` training sample.

Additional tables expand the description of the borrower and make the data set significantly more informative. The `bureau` table contains information about the client's past loans with other financial institutions, and `bureau_balance` reflects the monthly status of these loans. The `previous_application` table stores information about previous loan applications related to current customers. The `POS_CASH_balance` and `credit_card_balance` tables provide monthly cross-sections for previously issued loans and credit cards, and `installments_payments` records the history of actual payments, including missed ones. In other words, the dataset combines both static characteristics of the application and dynamic data that describe credit behavior over time. This makes the material especially valuable for scoring tasks, but at the same time significantly complicates its preparation.

Another important feature of the training sample is the pronounced class imbalance. In the `application_train` table, the value the `TARGET=0` occurs 282,686 times, while the value `TARGET=1` occurs only in 24,825 cases. In other words, the share of the positive class is about 8.07%. For a practical task, this means that most customers belong to a more prosperous group, and cases of default or serious payment difficulties are much less common. Such a structure is typical for credit scoring and requires a particularly careful choice of approaches to assessing the quality of models,

since high formal accuracy alone does not guarantee a good identification of risky borrowers.

Meaningfully, this dataset is a combination of several types of information. On the one hand, it contains the socio-demographic and financial characteristics of the current application. On the other hand, it includes information about the client's past credit activity, his interaction with financial institutions, and payment discipline for previously issued obligations. This makes it possible to analyze not only the borrower's current condition, but also his previous credit behavior. This is especially important for the scoring model, since the risk of default is usually determined not by one factor, but by a set of features, some of which are related to the client's current financial profile, and some to his history.

3.2 Primary data analysis

The main table contains 307,511 observations and 122 features. Each row corresponds to a single loan application, and the columns describe the client from different angles: income, loan parameters, marital status, employment, property characteristics, and other factors. Already at this stage, it was seen that the data has a mixed structure: some of the features are numeric, some are categorical, while some variables are not fully filled in.

The distribution of the target variable **TARGET** was considered separately. The analysis showed a pronounced class imbalance: most observations belong to **class 0**, while **class 1** is noticeably less common (Figure 3.2). The sample contains 282,686 objects of **class 0** and 24,825 objects of **class 1**, which corresponds to the share of the positive class of about 8.07%. This situation is typical for credit scoring tasks, but it requires a careful approach to assessing the quality of models, since high accuracy in such conditions does not always mean good identification of risky borrowers.

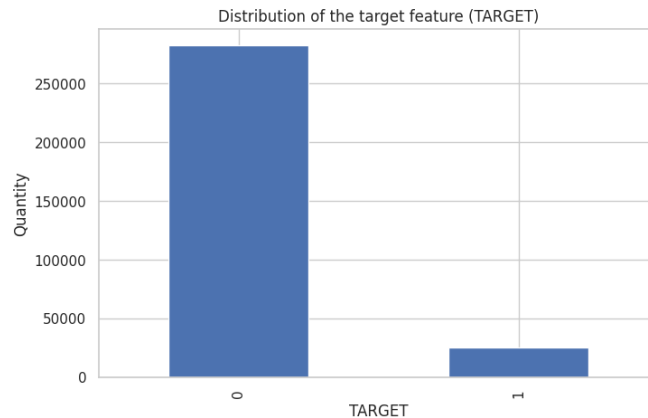


Figure 3.2: Distribution of the target variable (**TARGET**).

The next step was to analyze the gaps. It showed that there are a significant number of

features in the data with a high proportion of missing values. This is especially true for variables related to housing and real estate characteristics, such as `COMMONAREA_AVG`, `COMMONAREA_MODE`, `NONLIVINGAPARTMENTS_AVG`, `LIVINGAPARTMENTS_AVG` and a number of fields close to them. For some of these signs, the proportion of omissions turns out to be very high (Figure 3.3). This indicates the need for a separate strategy for processing missing values at the preprocessing stage.

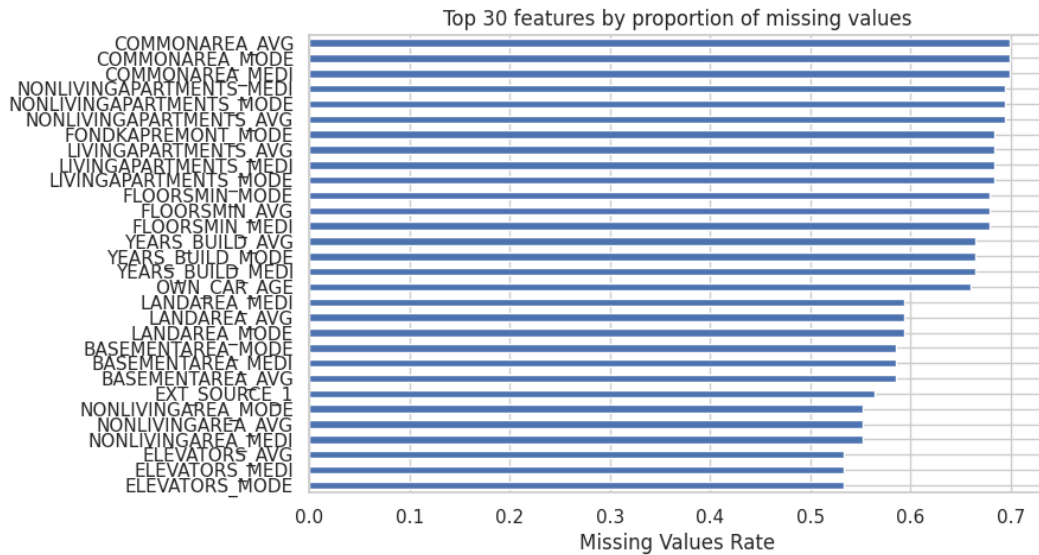


Figure 3.3: Features with the highest proportion of missing values.

A preliminary analysis of the numerical features was also performed. The Spearman correlation was used to evaluate their relationship with the target variable (Figures 3.4, 3.5). The most noticeable relationship with `TARGET` was shown by signs of the `EXT_SOURCE` family, as well as individual financial and demographic indicators. This suggests that already at the initial level, there are variables in the data that can give a strong signal for predicting the risk of default.

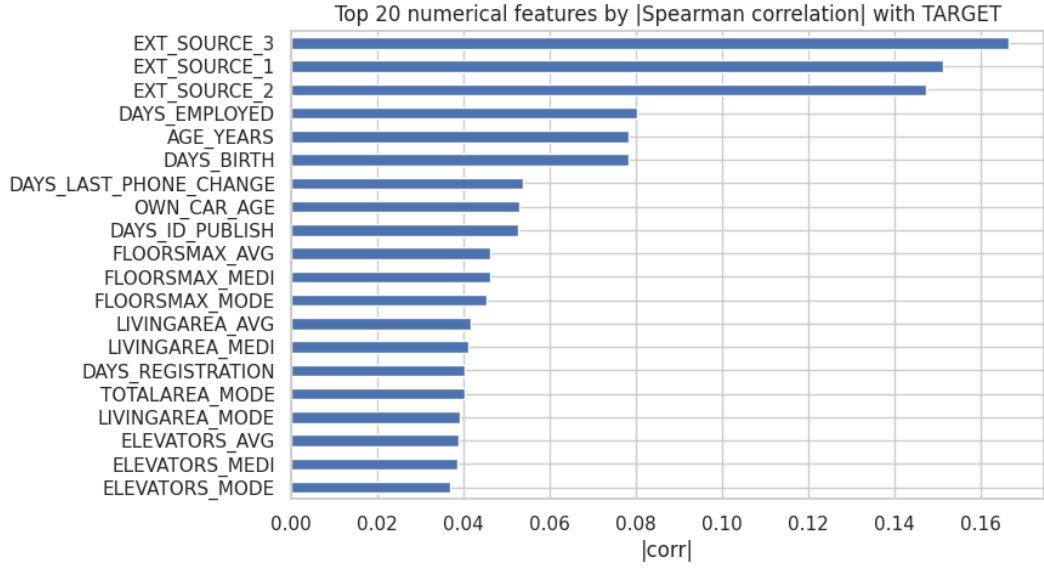


Figure 3.4: Numerical features with the highest modulo correlation with the target variable.

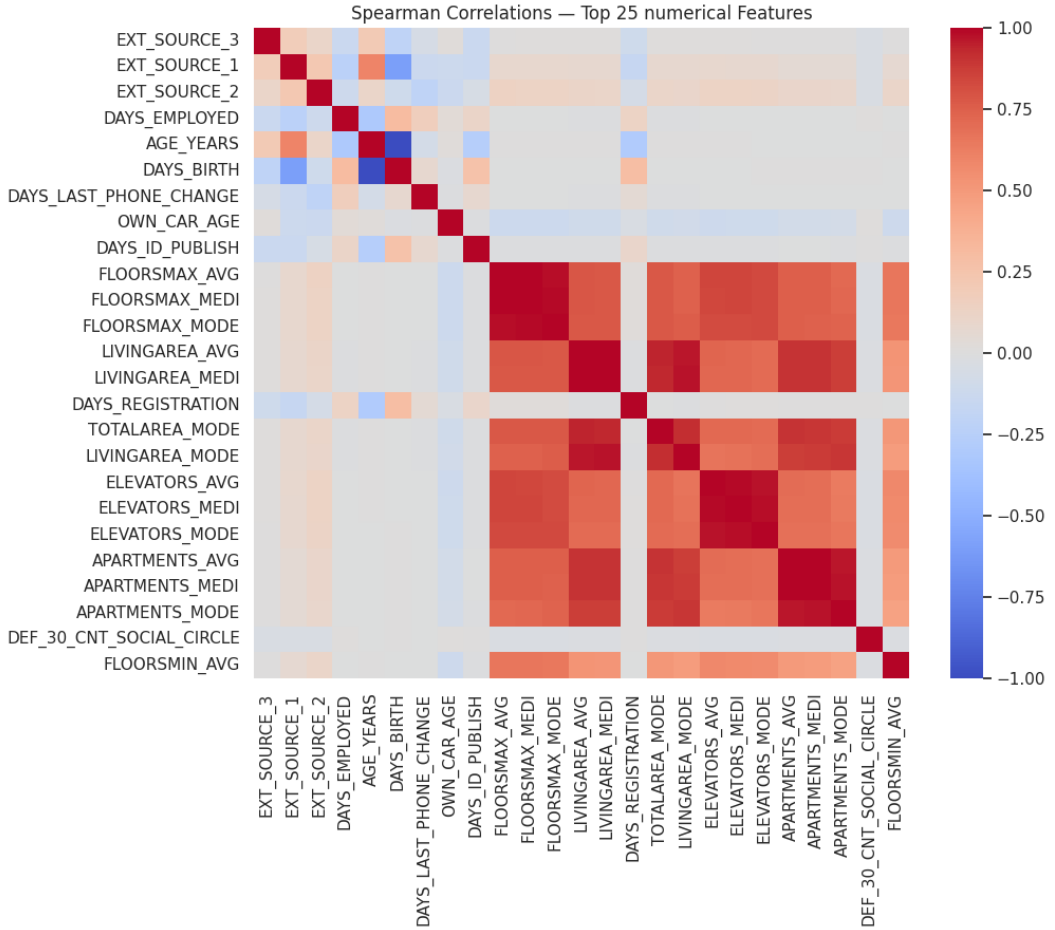


Figure 3.5: Heat map of correlations (top 25 numerical features according to $|corr|$).

As a result, it is possible to say that the training table has a large volume and a mixed structure of features. The target variable is noticeably unbalanced, and there are a significant number of gaps in a number of columns. At the same time, signs that are potentially important for predicting default were identified at the initial stage. All this confirms the need for further

data purification and transformation before building scoring models.

3.3 Data cleaning and transformation

At this step, special attention was paid to omissions, abnormal values, and the creation of derived features that could more accurately reflect the client's financial situation. The missing values were processed first. Since both numerical and categorical features were present in the dataset, different approaches were used for them. Numerical variables were filled in based on stable statistics, and categorical variables were filled in with a separate value that allowed saving information that the original field was not filled in. This approach is convenient for scoring tasks, since the absence of a value in some cases can be a useful signal in itself and should certainly not be lost during preprocessing.

Anomalous values were considered separately. There were signs in the data for which individual values were clearly out of line with the general logic and could be related not to the real features of the client, but to the way information was stored. Such cases required careful adjustments, because without this they could distort the resulting dependencies and reduce the quality of the models. At this stage, such values have been adjusted to a more correct form or taken into account as omissions for subsequent processing. In addition to cleaning the data, the transformation of part of the features was performed. To increase the information content of the set, derivative variables were formed that reflect the relationships between key financial characteristics. In particular, the criteria related to the ratio of the loan amount to income, the size of the annuity relative to income, as well as the share of the loan in relation to the value of the purchased property were considered. Such indicators make it possible to better describe the client's debt burden than the initial absolute values, and therefore often prove useful in credit scoring tasks.

As a result of the transformations performed, the data structure has become more convenient for further work. After adding new factors, the number of features increased from 122 to 126, which is precisely due to the formation of additional derived variables. At the same time, the data set retained its original content, but it became better prepared for the next stage, where it was necessary to separately identify numerical and categorical features.

3.4 Identification of numerical and categorical features

At this stage, the service columns that are not directly involved in training the model were first separated from the data set. These fields include the application identifier, which

is needed for communication and accounting of observations, but does not contain meaningful information about credit risk. The target variable was also not included in the features, as it was used separately as the correct answer for training and evaluating models. The remaining feature space was divided by data types. As a result, after all the preliminary transformations, 74 numerical and 51 categorical features were identified. This ratio shows that both groups of variables are represented in a significant amount in the data set, which means that the final quality of the model will largely depend not only on the chosen algorithm, but also on how well each of these groups is processed. Variables expressed as quantitative values were classified as numerical features. This group includes, for example, the client’s income, loan amount, annuity amount, age, employment experience, as well as derivative financial indicators generated at the previous stage. Such variables directly reflect the measurable characteristics of the borrower and allow the model to identify the relationship between the level of financial burden, the socio-economic situation of the client and the probability of default.

Categorical features, in turn, describe the client’s belonging to a certain group. These include gender, type of employment, marital status, level of education, housing conditions, type of contract, and other similar characteristics. Such signs contain important information, but most machine learning algorithms cannot work with them directly in text form. For this reason, their separation into a separate group was necessary not only from a technical point of view, but also for the subsequent comparison of different encoding methods, which is discussed below.

The separation of features by type made it possible to make further processing more orderly. For numeric variables, it has become possible to use separate preprocessing steps related to gap filling and scaling. For categorical features, on the contrary, the emphasis was placed on choosing the appropriate encoder.

3.5 Preprocessor construction

The preprocessor was implemented in such a way as to process numeric and categorical variables separately. For numerical features, filling in the gaps with the median was used. This method is well suited for tabular data with possible outliers, since the median is a more stable characteristic than the average value. After that, scaling using `StandardScaler` was additionally applied to some of the models. First of all, this applied to logistic regression, for which reducing features to a comparable scale allows for more stable learning and a more correct interpretation of coefficients.

For categorical features, the preprocessor also filled in the gaps first. At this stage, the

substitution of the most common value was used. After that, coding of categorical variables was applied. Since one of the tasks of the work was to compare different approaches to the representation of categorical features, the preprocessor was built not for one fixed encoding method, but for several variants at once. The project included four coding strategies for categorical variables: `One-Hot Encoding`, `Ordinary Encoding`, `Target Encoding`, and `CatBoost Encoding`. For `One-Hot Encoding`, each category was converted into a separate binary feature. This option is well suited for interpreted models and allows explicit consideration of individual values of categorical variables. When using `Ordinary Encoding`, numeric codes were assigned to categories, which makes the representation of features more compact, but requires caution in interpretation. `Target Encoding` and `CatBoost Encoding` allowed us to take into account the relationship of a category with a target variable, which is especially useful for tabular scoring data with a large number of categorical features.

Technically, the preprocessor was implemented through `ColumnTransformer`, which combined two independent processing branches: one for numeric variables, the other for categorical ones. This made it possible to apply different operations to different types of data within the same pipeline, without mixing them with each other. This approach is convenient not only from the point of view of software implementation, but also from the point of view of project logic: each type of feature is processed according to its nature. An important advantage of the built preprocessor was its versatility. The same common framework could be used when comparing different encoders and different models without having to manually rebuild the entire data preparation process. Due to this, it was possible to make the subsequent experiment more reproducible and visual. All the changes concerned only individual pipeline elements, such as the type of encoder or model, while the basic data processing scheme remained the same. In addition, the use of a single preprocessor made it possible to avoid inconsistencies between the training and testing stages. All feature transformations were performed in the same way for both the training and the verification part of the sample. For a practical task, this is of fundamental importance, since any discrepancies in data processing can lead to distortion of the results and reduce the reliability of the model.

3.6 Comparison of encoders and models

For comparison, four approaches to encoding categorical variables were chosen: `One-Hot Encoding`, `Ordinary Encoding`, `Target Encoding`, and `CatBoost Encoding`. Each of them was tested in combination with two models — `logistic regression` and `LightGBM`. This comparison

allowed us to evaluate not only the influence of the encoder itself, but also how differently the linear and boosting models react to the same way of representing features. The quality assessment was carried out using stratified cross-validation. ROC-AUC, Average Precision, LogLoss, and Accuracy were considered as the main metrics. This set of indicators was not chosen by chance. In conditions of class imbalance, accuracy alone is not enough, so the main focus was on ROC-AUC and Average Precision, which better reflect the model's ability to distinguish between reliable and risky borrowers.

The results showed that the LightGBM model demonstrated the best quality in the comparison under consideration. In all coding variants, it was significantly ahead of the logistic regression in terms of ROC-AUC and Average Precision. The best ROC-AUC result was obtained for the combination of Target Encoding + LightGBM, where the metric value was 0.7595. Ordinary Encoding + LightGBM and One-Hot Encoding + LightGBM also showed very similar results, which indicates a fairly high stability of the boosting model to the choice of a specific method for encoding categorical features.

For logistic regression, One-Hot Encoding turned out to be the best option. In this case, the ROC-AUC value was 0.7442, which is higher than when using Ordinary, Target, and CatBoost Encoding. This picture looks logical. For a linear model, the way categorical features are represented is especially important, since it depends on how well the differences between individual categories are taken into account. In this sense, One-Hot Encoding turns out to be a more natural option, since it does not introduce an artificial order between the values of the feature.

If we compare the results within each model, we can see that the difference between LightGBM encoders remains relatively small. This means that the model itself adapts quite well to different ways of representing categorical variables. In logistic regression, the differences are more pronounced, and the choice of an encoder has a more noticeable effect on the final metrics. Therefore, for linear interpretable models, the issue of encoding categorical features is particularly important.

From the point of view of LogLoss, the results were consistent with the conclusions on ROC-AUC. The lowest LogLoss value was also obtained for LightGBM, primarily in combination with Target Encoding. This means that this combination not only separates classes better, but also provides better probabilistic estimates. At the same time, the Accuracy values for all the considered combinations turned out to be close to each other and were at a high level. However, in this case, this should not be misleading, since with an imbalance of classes, high accuracy is achieved relatively easily and does not allow us to judge the real usefulness of the model by itself.

In terms of content, the results obtained allow us to make several observations(Figure 3.6). Firstly, already at the stage of preliminary comparison, it became clear that a more complex model based on tabular data is capable of yielding quality gains even with a close preprocessing scheme. Secondly, the influence of the encoder is indeed present, but it manifests itself in different ways depending on the chosen algorithm. Thirdly, for further work, it makes sense to consider not one universal encoding method, but several options, since the best choice depends on the model and on what balance between quality and interpretability is required.

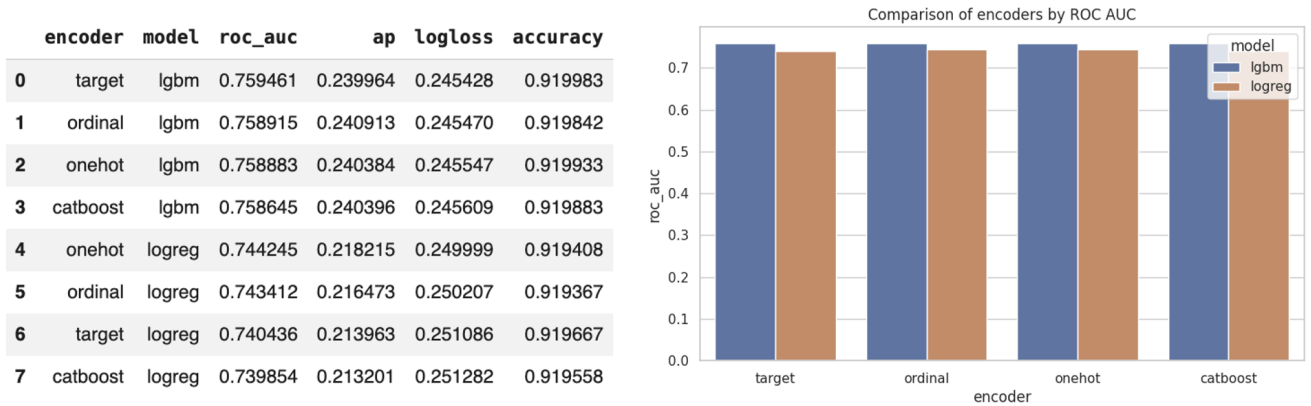


Figure 3.6: Results of comparing encoders and models using ROC-AUC, Average Precision, LogLoss, and Accuracy metrics.

4 Development of credit scoring models and their interpretation

4.1 Selection of informative features

Feature selection was performed using several methods. First, `SelectKBest` with mutual information was applied as a fast preliminary filter. It retained 50 features and, together with `LightGBM`, gave $\text{ROC-AUC} = 0.7218$, $\text{Average Precision} = 0.1956$, $\text{LogLoss} = 0.2554$, and $\text{Accuracy} = 0.9191$. This showed that the model kept acceptable quality even after reducing the feature space.

The main focus was then placed on `Permutation Importance`, since it reflects the actual contribution of features after model training and preserves interpretation at the level of the original variables. The most important features were `EXT_SOURCE_3`, `EXT_SOURCE_2`, `CREDIT_ANNUITY_RATIO`, `EXT_SOURCE_1`, `AMT_GOODS_PRICE`, `AMT_CREDIT`, `AMT_ANNUITY`, `CODE_GENDER`, `NAME_EDUCATION_TYPE`, `NAME_CONTRACT_TYPE`, `DAYS_BIRTH`, `AGE_YEARS`, `NAME_FAMILY_STATUS`, `DAYS_ID_PUBLISH`, and `DAYS_EMPLOYED`. These results show that the model relies mainly on external scoring indicators, financial application characteristics, and demographic and social features (Figure 4.1).

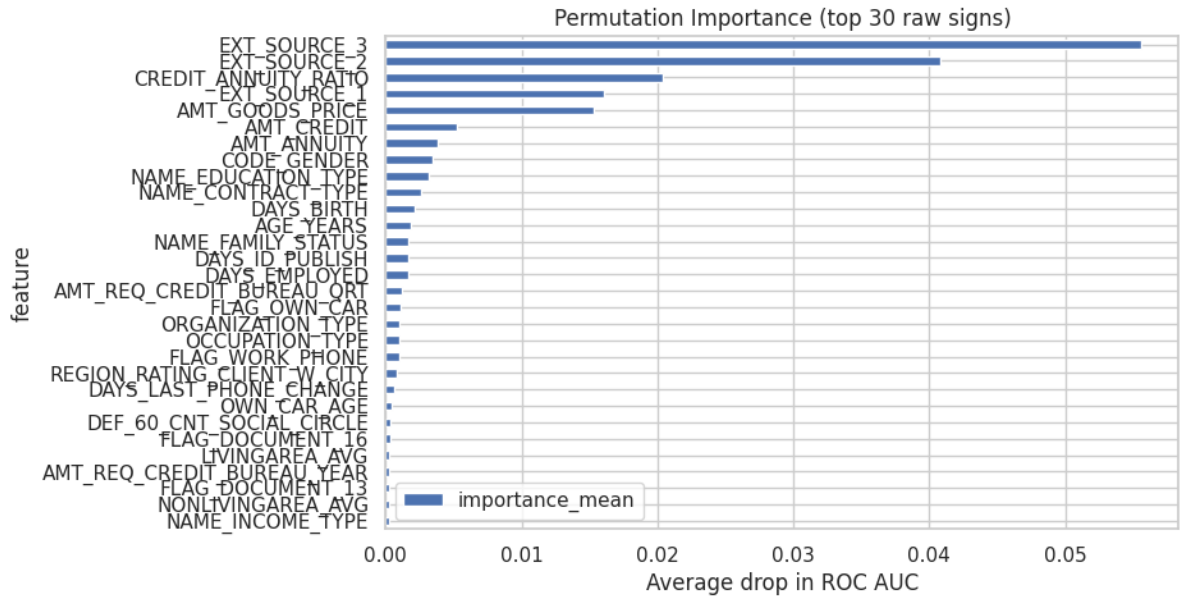


Figure 4.1: Top 30 signs of Permutation Importance.

Additionally, a SHAP-based approach was used to check whether the results were consistent with `Permutation Importance`. The analysis again highlighted the `EXT_SOURCE` features, application-related financial variables, and indicators of age and debt burden as the most important factors. This showed that the key risk drivers were stable and did not depend on a single

feature selection method (Figure 4.2).

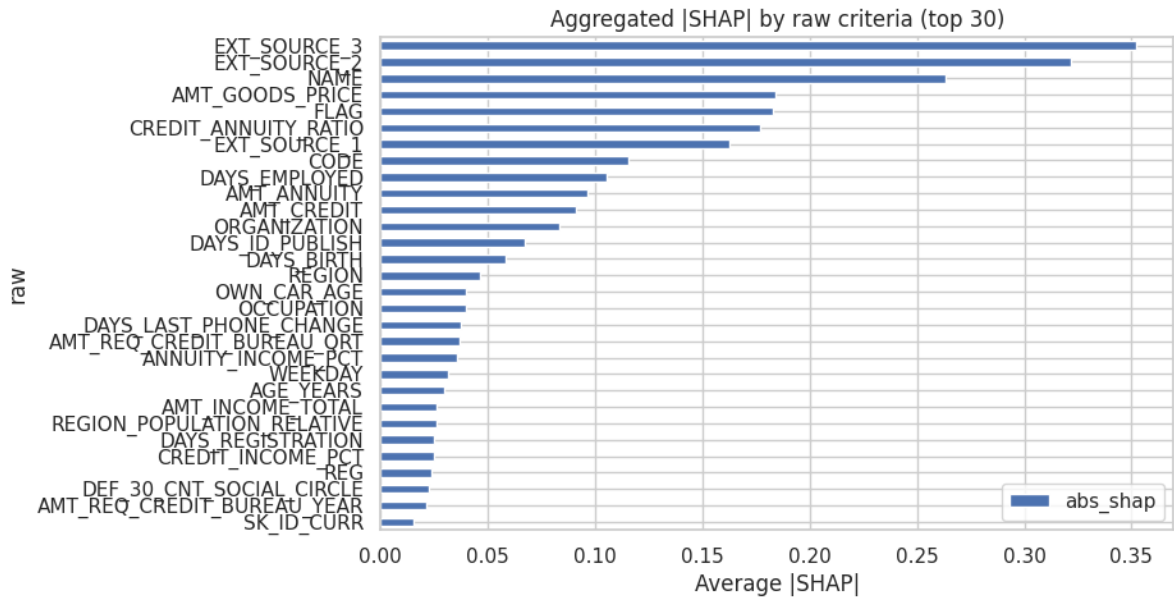


Figure 4.2: Aggregated importance of features by SHAP.

Based on the results, a final set of 50 initial features was selected for future modelling, including 30 numeric and 20 categorical variables. Although this set was much smaller than the original feature space, it preserved the most important information for forecasting and was used to build the logistic regression, decision tree, and LightGBM models. Feature selection solved two tasks at once: it reduced redundancy in the input data and improved interpretability. A more compact and meaningful set of variables made it easier to explain which client characteristics influenced the final risk assessment and in what direction.

4.2 Logistic regression model

The choice of this algorithm is justified by the fact that logistic regression remains one of the most interpretable and practically convenient models for credit scoring. It provides probabilistic predictions and makes it possible to analyze how features influence default risk.

The model was trained on the selected set of 50 features, including 30 numerical and 20 categorical variables. The pipeline used imputation and scaling for numerical features, and imputation with **One-Hot Encoding** for categorical ones. This encoding method was chosen because it showed the best results for logistic regression at the preliminary comparison stage. Model quality was evaluated on a holdout sample using **ROC-AUC**, **Average Precision**, **LogLoss**, and **Accuracy**. The results showed that logistic regression provides stable and sufficiently strong performance: **ROC-AUC** = 0.7480, **Average Precision** = 0.2301, **LogLoss** = 0.2491, and **Accuracy** = 0.9195

(Figure 4.3). Although these values were not the highest among all models, they were quite convincing for an interpretable linear approach. In this task, **ROC-AUC**, **Average Precision**, and **LogLoss** are more informative than accuracy, since class imbalance makes high accuracy easier to achieve.

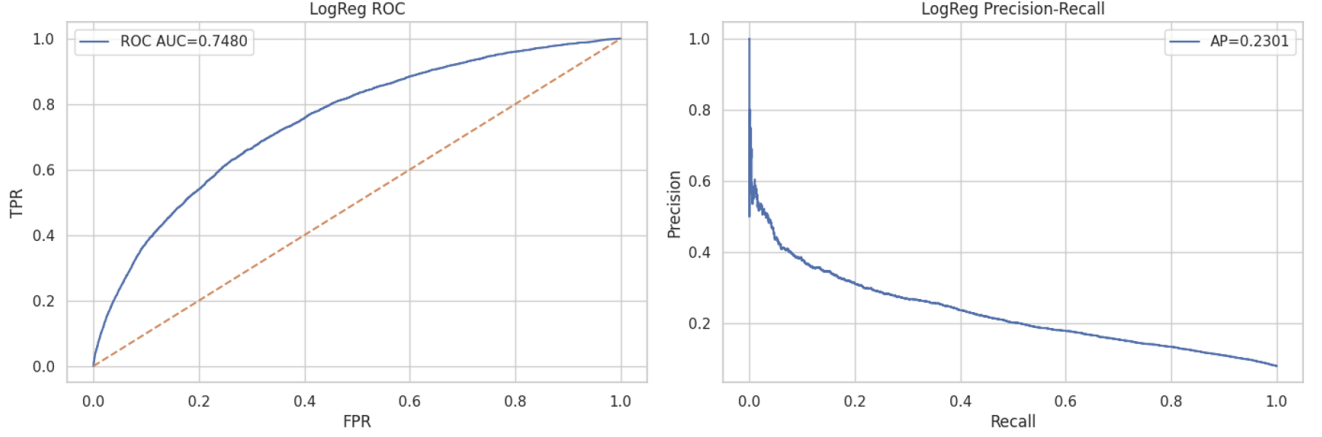


Figure 4.3: ROC curve and Precision–Recall curve for the logistic regression model.

A key advantage of logistic regression in this work is coefficient interpretability. The sign of a coefficient shows whether a feature increases or decreases the predicted risk, while its magnitude reflects relative importance after preprocessing. The analysis showed that the strongest effects came from `EXT_SOURCE_2`, `EXT_SOURCE_3`, and features related to loan parameters and debt burden, such as `AMT_CREDIT`, `AMT_GOODS_PRICE`, `AMT_ANNUITY`, and the loan-to-annuity ratio (Figure 4.4). This means that the model relies mainly on external scoring indicators, financial application characteristics, and some demographic and social variables, which makes its predictions both technically and practically interpretable.

At the same time, logistic regression retains its understandable limitations. The linear model does not take into account complex nonlinear dependencies and interactions between features better than boosting algorithms. For this reason, in terms of final quality, it is inferior to LightGBM, which was considered earlier in the comparison of models. However, within the framework of this work, this does not reduce the value of logistic regression. On the contrary, it acts as an important basic guideline: it allows us to obtain a clear, stable and sufficiently strong quality model, with which more complex approaches can then be compared.

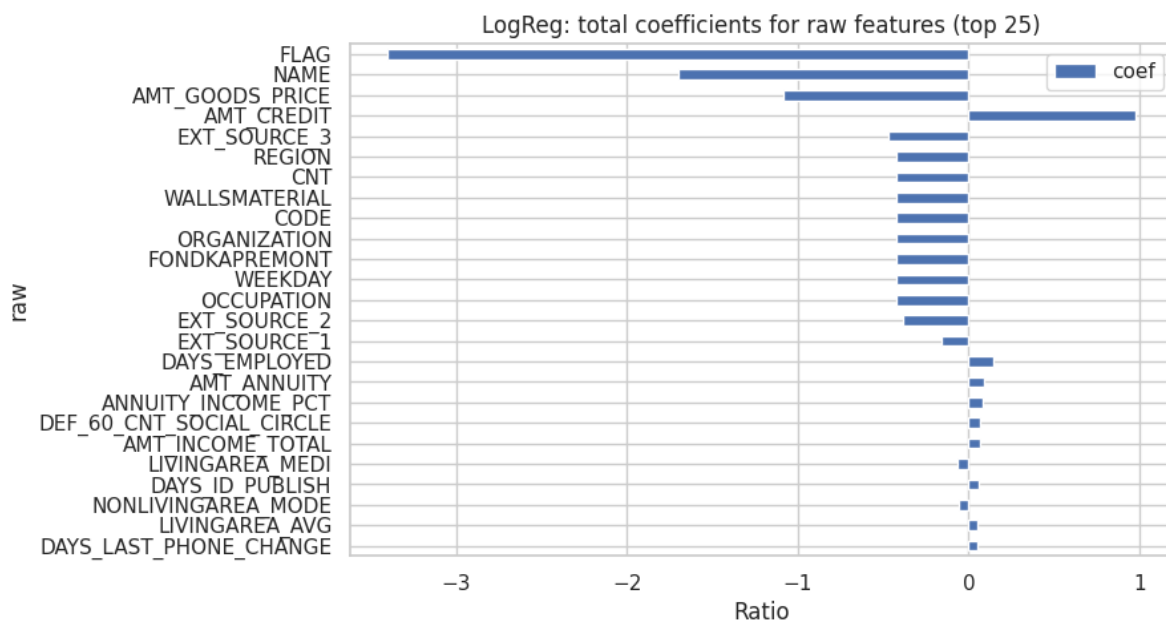


Figure 4.4: The most significant coefficients of the logistic regression model.

4.3 Decision tree

The next step was to build a decision tree. This algorithm is useful for credit scoring because it provides not only predictions, but also clear decision rules. Unlike logistic regression, it can capture non-linear relationships between features while still remaining interpretable, especially when tree depth is limited.

The model was trained on the same 50 selected features as logistic regression, including 30 numerical and 20 categorical variables. **One-Hot Encoding** was used for categorical features, while numerical features were imputed without additional scaling. To preserve interpretability and avoid overfitting, the tree was built with restricted parameters: `max_depth = 4` and `min_samples_leaf = 0.01`. The model was evaluated on a holdout sample using **ROC-AUC**, **Average Precision**, **LogLoss**, and **Accuracy**. The results were **ROC-AUC** = 0.7018, **Average Precision** = 0.1757, **LogLoss** = 0.2595, and **Accuracy** = 0.9193 (Figure 4.5). These values were lower than those of logistic regression and LightGBM, but they were expected for a shallow interpretable tree. In this case, the decision tree was used mainly as a transparent model that makes the prediction process easier to explain.

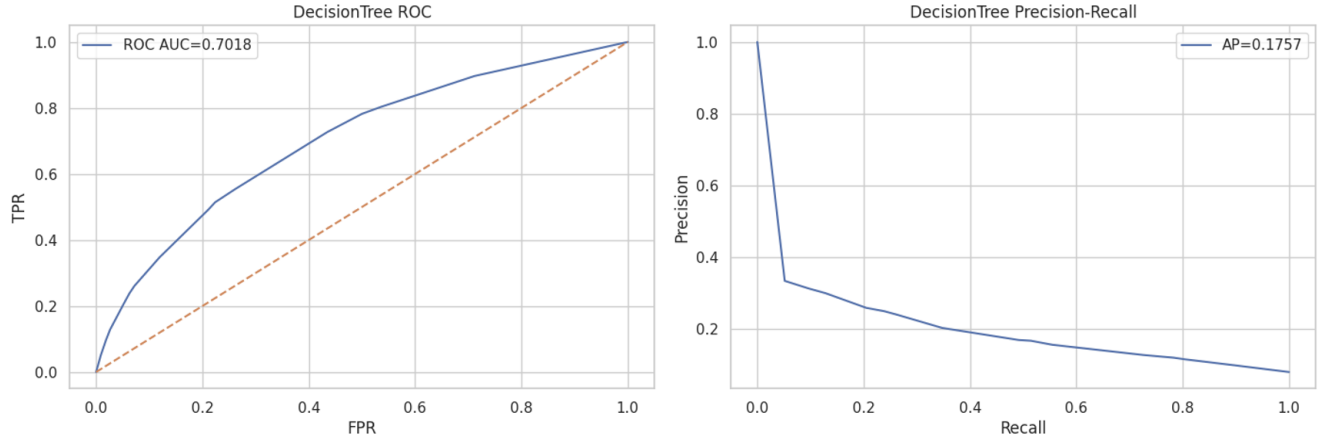


Figure 4.5: ROC curve and Precision-Recall curve for the decision tree model.

An important advantage of the decision tree is that its rules can be examined directly [1]. The analysis showed that `EXT_SOURCE_3`, `EXT_SOURCE_2`, and `EXT_SOURCE_1` already play a key role at the upper levels of the tree. This confirms the earlier conclusion that external scoring indicators are among the strongest predictors of default risk. The tree first separates clients by the ranges of these external scores and then refines the decision using additional features (Figure 4.6).

```

|--- EXT_SOURCE_3 <= 0.31
|   |--- EXT_SOURCE_2 <= 0.35
|   |   |--- EXT_SOURCE_3 <= 0.16
|   |   |   |--- class: 0
|   |   |   |--- EXT_SOURCE_3 > 0.16
|   |   |       |--- EXT_SOURCE_2 <= 0.21
|   |   |       |   |--- class: 0
|   |   |       |   |--- EXT_SOURCE_2 > 0.21
|   |   |       |   |--- class: 0
|   |   |--- EXT_SOURCE_2 > 0.35
|   |       |--- EXT_SOURCE_3 <= 0.13
|   |       |   |--- EXT_SOURCE_2 <= 0.59
|   |       |   |   |--- class: 0
|   |       |   |   |--- EXT_SOURCE_2 > 0.59
|   |       |   |   |--- class: 0
|   |       |--- EXT_SOURCE_3 > 0.13
|   |       |   |--- EXT_SOURCE_2 <= 0.62
|   |       |   |   |--- class: 0
|   |       |   |   |--- EXT_SOURCE_2 > 0.62
|   |       |   |   |--- class: 0

```

Figure 4.6: A fragment of decision tree rules.

An additional advantage of the decision tree is the direct interpretability of its rules. The analysis showed that `EXT_SOURCE_3`, `EXT_SOURCE_2`, and `EXT_SOURCE_1` were again the most important features, together with several financial application characteristics. This confirms a consistent pattern across models: external scoring indicators and variables related to credit burden make the strongest contribution to the forecast.

From a practical perspective, the main strength of the decision tree is its clarity. It makes it possible to trace the path from the input features to the final prediction, which is useful for explaining the model. At the same time, the tree is sensitive to changes in the data, and even with limited depth its structure may vary across different sample splits, while predictive quality

usually remains below that of stronger ensemble methods.

4.4 LightGBM

The next step was to build the **LightGBM** model. This algorithm is well suited for tabular data and often outperforms linear and single-tree approaches because it can capture complex nonlinear relationships between features. The model was trained on the same 50 selected features as the previous algorithms. **One-Hot Encoding** was used for categorical variables, while missing values in numerical features were imputed. A moderate configuration was chosen to balance predictive quality and stability.

On the holdout sample, the model achieved $\text{ROC-AUC} = 0.7711$, $\text{Average Precision} = 0.2630$, $\text{LogLoss} = 0.2420$, and $\text{Accuracy} = 0.9199$. Among all the considered models, **LightGBM** showed the best performance. The most informative metrics here were **ROC-AUC** and **Average Precision**, since they better reflect the model's ability to distinguish between reliable and risky borrowers under class imbalance (Figure 4.7).

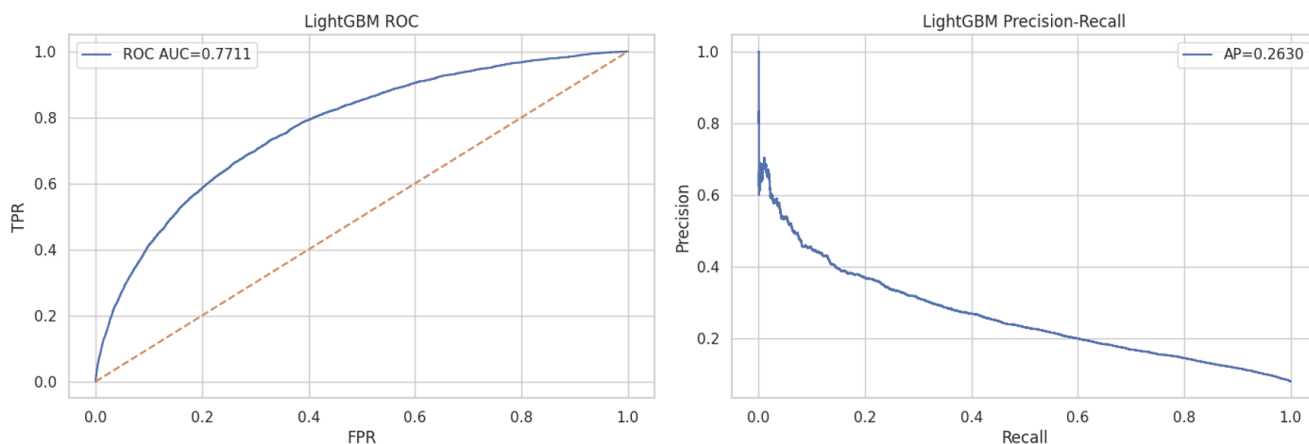


Figure 4.7: ROC curve and Precision-Recall curve for the LightGBM model.

The result is consistent with expectations. Unlike logistic regression, **LightGBM** can capture nonlinear relationships, and unlike a single decision tree, it uses an ensemble of trees, which leads to stronger predictions. Feature importance analysis showed that the most significant variables were `EXT_SOURCE_3`, `EXT_SOURCE_2`, `CREDIT_ANNUITY_RATIO`, `EXT_SOURCE_1`, `DAYS_EMPLOYED`, `AMT_GOODS_PRICE`, `DAYS_BIRTH`, `AMT_ANNUITY`, `AGE_YEARS`, and `AMT_CREDIT`. This shows that the model mainly relies on external scoring indicators, debt burden, loan parameters, and demographic characteristics (Figure 4.8).

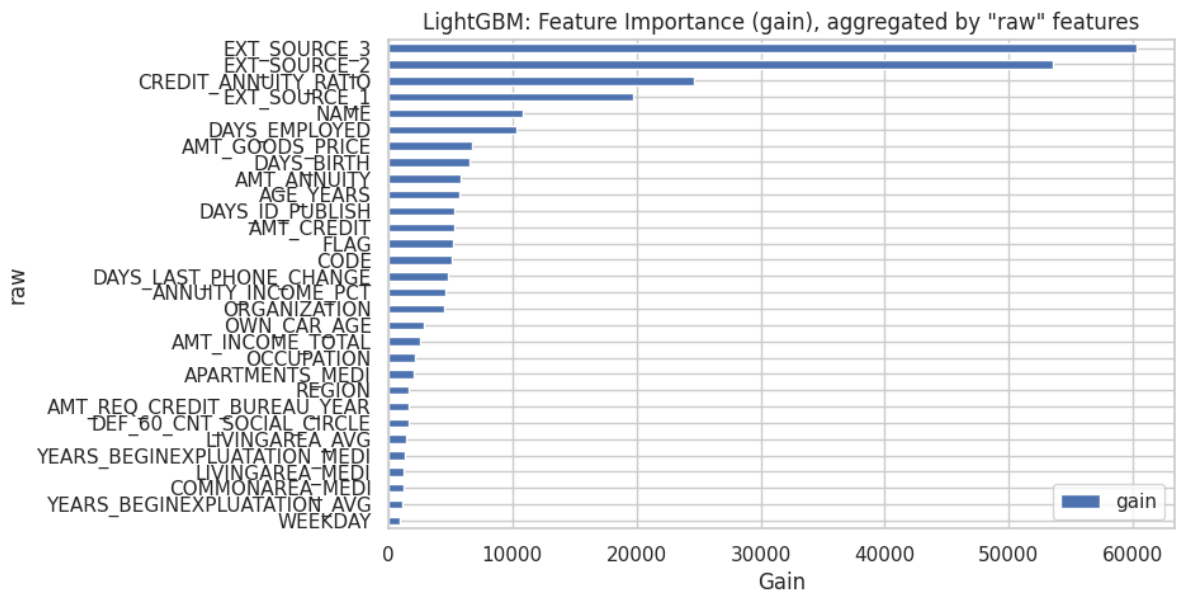


Figure 4.8: The importance of features in the gain LightGBM model.

Additionally, partial dependence plots were built for the most important features. They showed that higher values of `EXT_SOURCE_2` and `EXT_SOURCE_3` were associated with lower predicted risk, while a higher `CREDIT_ANNUIITY_RATIO` increased the probability of default. These relationships were consistent and supported the interpretability of the model.

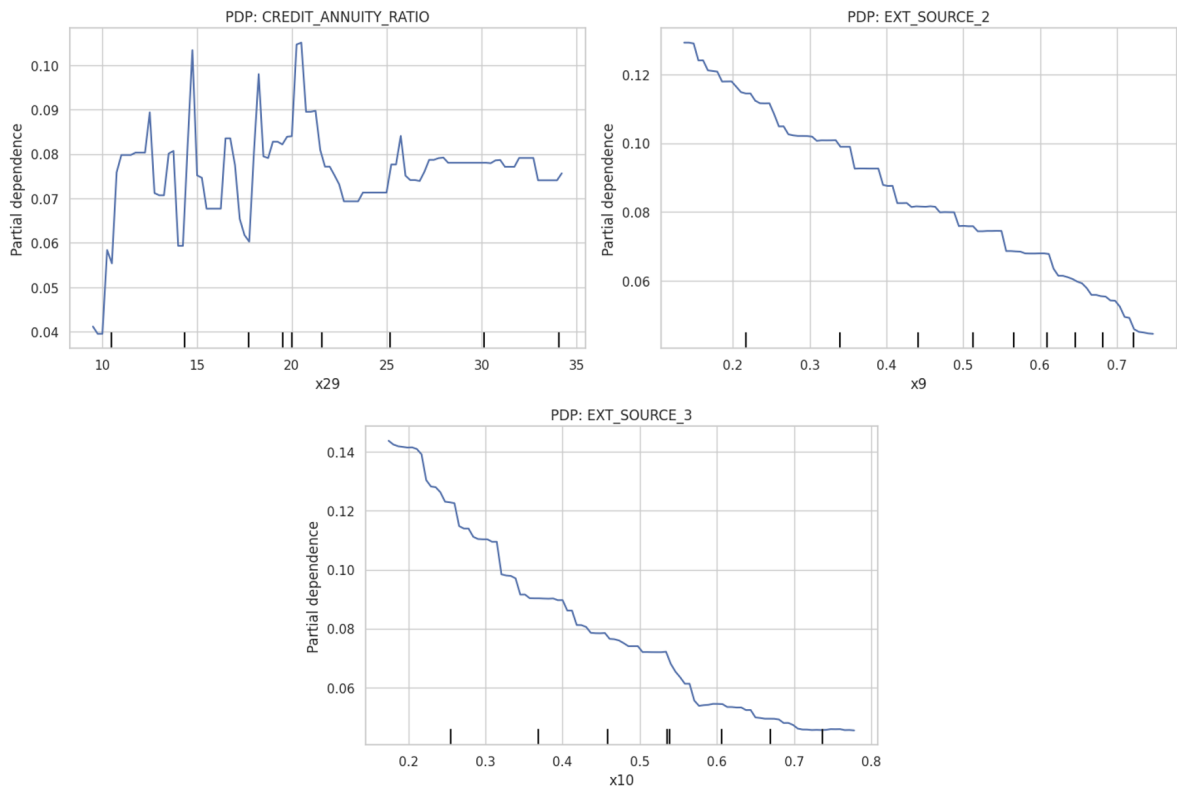


Figure 4.9: Partial dependency graphs for `EXT_SOURCE_3`, `EXT_SOURCE_2` and `CREDIT_ANNUIITY_RATIO`.

4.5 MLP Small and MLP Deep models

In addition to classical models and gradient boosting, two neural network models were considered: MLP Small and MLP Deep. The purpose of this stage was to check whether multilayer perceptrons can improve the quality of credit scoring by capturing nonlinear dependencies in the selected feature space.

MLP Small was used as a compact neural baseline. It contained one hidden layer with 64 neurons, ReLU activation, Adam optimizer, and early stopping. MLP Deep had a more complex architecture with two hidden layers of 128 and 64 neurons. It was trained with the same basic approach, but with a deeper structure and stronger ability to model nonlinear relationships. Both models were included in the same general preprocessing pipeline as the previous algorithms. Numerical features were imputed and scaled, while categorical features were encoded using the methods considered earlier.

At the preliminary stage, the models were compared with One-Hot Encoding, Ordinal Encoding, Target Encoding, and CatBoost Encoding. The results showed that neural networks were more sensitive to the choice of categorical encoding than LightGBM. For MLP Deep, the best cross-validation result was obtained with One-Hot Encoding: `ROC-AUC` = 0.7253, `Average Precision` = 0.1969, `LogLoss` = 0.2568, and `Accuracy` = 0.9201. For MLP Small, the best `ROC-AUC` was achieved with `Ordinal Encoding` and amounted to 0.7091. This means that the quality of neural models depended noticeably on the way categorical variables were represented. After that, both models were trained on the final set of 50 selected features, including 30 numerical and 20 categorical variables. On the holdout sample, MLP Small achieved `ROC-AUC` = 0.7505, `Average Precision` = 0.2344, `LogLoss` = 0.2490, and `Accuracy` = 0.9194. MLP Deep showed a slightly better result: `ROC-AUC` = 0.7514, `Average Precision` = 0.2354, `LogLoss` = 0.2483, and `Accuracy` = 0.9196. Thus, both neural networks slightly outperformed logistic regression, but remained weaker than LightGBM.

The results show that MLP models were able to capture part of the nonlinear structure of the data. However, the gain compared to logistic regression was small, and the deeper architecture did not provide a significant improvement over the compact model. From a practical point of view, this is important because neural networks are less transparent than logistic regression and decision trees. Therefore, in this work MLP Small and MLP Deep should be considered as additional nonlinear benchmarks rather than as the main scoring models.

4.6 Comparison of models by quality

The models were compared using ROC-AUC, Average Precision, LogLoss, and Accuracy. This set of metrics provides a balanced view of model quality: ROC-AUC reflects class separation, Average Precision is especially informative under class imbalance, LogLoss evaluates probabilistic predictions, and Accuracy shows the share of correct classifications.

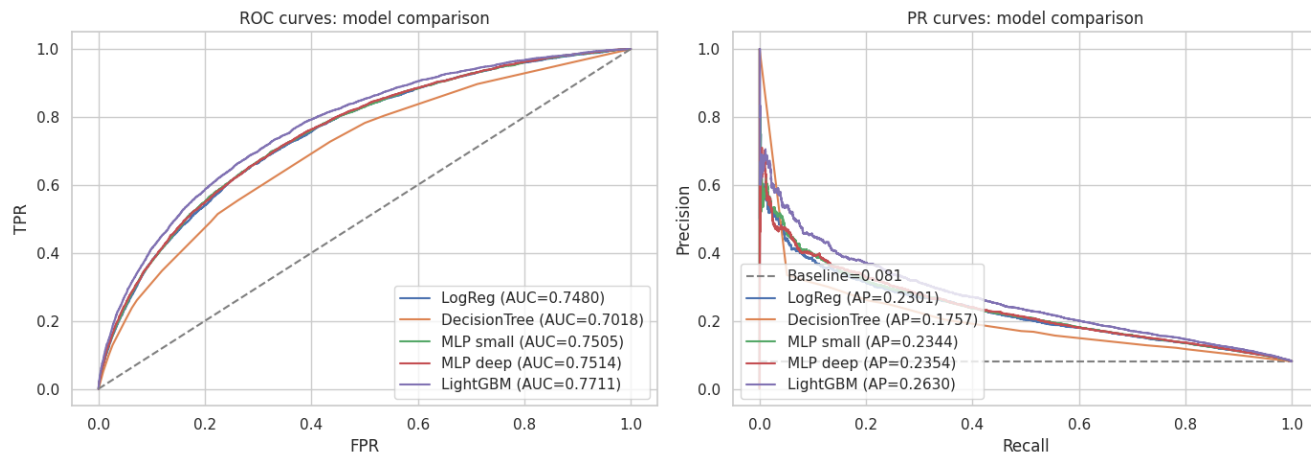


Figure 4.10: ROC and Precision–Recall curves for the comparison of logistic regression, decision tree, MLP Small, MLP Deep, and LightGBM models

The results showed that **LightGBM** performed best, with ROC-AUC = 0.7711, Average Precision = 0.2630, LogLoss = 0.2420, and Accuracy = 0.9199. Logistic regression took an intermediate position, achieving ROC-AUC = 0.7480, Average Precision = 0.2301, LogLoss = 0.2491, and Accuracy = 0.9195. The decision tree showed the weakest quality, with ROC-AUC = 0.7018, Average Precision = 0.1757, LogLoss = 0.2595, and Accuracy = 0.9193. The Accuracy values were similar for all three models, around 0.92, so this metric was not the most informative because of class imbalance. The main differences appeared in ROC-AUC, Average Precision, and LogLoss. Overall, **LightGBM** provided the best predictive quality, logistic regression offered a good balance between quality and interpretability, and the decision tree remained the most transparent but the least accurate model.

4.7 Interpretation of modeling results

The interpretation combined several approaches: coefficient analysis for logistic regression, rule analysis for the decision tree, and feature importance, SHAP values, and partial dependence plots for **LightGBM**. This made it possible to compare the results across models and identify the most consistent risk factors.

The analysis showed a stable overall pattern. In logistic regression, decision tree, and

LightGBM, the most important features included EXT_SOURCE_2, EXT_SOURCE_3, EXT_SOURCE_1, CREDIT_ANNUITY_RATIO, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE and DAYS_BIRTH. This suggests that the forecast mainly depends on external scoring indicators, loan parameters, debt burden, and some demographic characteristics.

In logistic regression, higher values of EXT_SOURCE_2 and EXT_SOURCE_3 reduced predicted risk, while greater credit burden increased it. The decision tree reflected the same pattern through sequential rules, with EXT_SOURCE features appearing at the top splits. For LightGBM, SHAP values and partial dependence plots confirmed that higher EXT_SOURCE_2 and EXT_SOURCE_3 lowered risk, whereas a higher CREDIT_ANNUITY_RATIO increased the probability of default.

The neural network models, MLP Small and MLP Deep, were less transparent than the previous models. They slightly improved the quality compared to logistic regression, but their internal decision mechanism could not be interpreted directly through coefficients or simple decision rules. Therefore, in this work they were used mainly as additional nonlinear benchmarks. Their results confirmed that nonlinear dependencies are present in the data, but they did not provide a clearer explanation of credit risk factors than logistic regression, decision tree, or LightGBM.

Overall, the interpretation showed that the main models relied on clear and meaningful risk factors. LightGBM provided the best predictive quality, logistic regression offered the clearest coefficient-based interpretation, and the decision tree gave the most transparent rule-based explanation. The MLP models occupied an intermediate position in terms of quality, but were less suitable for direct interpretation in the context of explainable credit scoring.

5 Checking the stability of models and features

5.1 Setting the stability assessment task

Model stability was evaluated because machine learning algorithms may react differently to small changes in the data. If quality metrics, feature importance, or predictions vary too much across runs, the practical value of the model decreases. In this work, stability was assessed by repeating experiments on different data splits. Three aspects were considered: the stability of quality metrics, the stability of feature importance, and the stability of predictions. This made it possible to treat stability not as a separate technical check, but as an important characteristic of model reliability.

As a result, the models were evaluated not only by their performance on a single split, but also by how consistently they retained their properties across repeated runs.

5.2 Experiment with different splits and random initializations

To test model stability, a separate experiment was conducted with repeated runs on different data splits. Three models were considered: logistic regression, decision tree, and `LightGBM`. All of them used the same preprocessing scheme, which made it possible to compare the stability of the models themselves rather than the effect of different data preparation methods. To reduce computational cost, the analysis was performed on a subsample of 5.000 observations. Five runs were carried out with different `random_state` values and new stratified splits. Within each run, 5-fold cross-validation was applied.

For each model, quality metrics, feature importance values, and probabilistic predictions were recorded. This made it possible to assess stability from several perspectives at once: forecast quality, consistency of important features, and reproducibility of predictions.

5.3 Analysis of the stability of quality metrics

After repeated runs, the stability of quality metrics was analyzed for logistic regression, decision tree, and `LightGBM`. The main goal was to evaluate how strongly the results changed across different data splits rather than to compare the models only by absolute quality.

The analysis used the mean values of ROC-AUC, LogLoss, TPR, and FPR across five runs, together with their standard deviations. Logistic regression showed the most stable behavior, with ROC-AUC = 0.7160 and a standard deviation of 0.0040, while LogLoss was 0.2690 with a standard deviation of 0.002. The decision tree was the least stable: its mean ROC-AUC was 0.6485 with a standard deviation of 0.0083, and its LogLoss showed the largest spread, with a mean of 0.3178 and a standard deviation of 0.0211. `LightGBM` took an intermediate position, with ROC-AUC = 0.6940 and a standard deviation of 0.0044, while LogLoss was 0.2890 with a standard deviation of 0.0020.

A similar pattern was observed for TPR and FPR: logistic regression and `LightGBM` varied moderately, whereas the decision tree was more sensitive to changes in the data. Although the experiment was conducted on a reduced subsample, the results were sufficient to show that logistic regression was the most stable model, `LightGBM` also remained fairly consistent, and the decision tree was the least reliable in terms of metric reproducibility.

5.4 Analysis of the stability of the importance of features

For logistic regression and the decision tree, stability was assessed using permutation importance, while for `LightGBM`, gain importance and SHAP values were also analyzed. Spearman

correlation across runs was then used to measure how consistently the models ranked features by importance.

Logistic regression showed the highest stability: correlations were around 0.9 or higher, and the set of top features remained almost unchanged (Figures 5.1, 5.2). This indicates that the model not only produced stable quality metrics, but also identified key risk factors in a reproducible way, including `EXT_SOURCE` features, credit burden indicators, and some demographic variables.

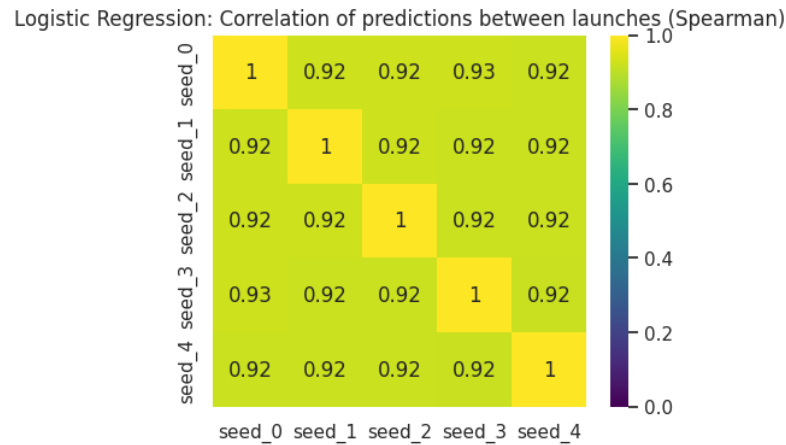


Figure 5.1: Heatmap of correlation of importance of logistic regression between launches.

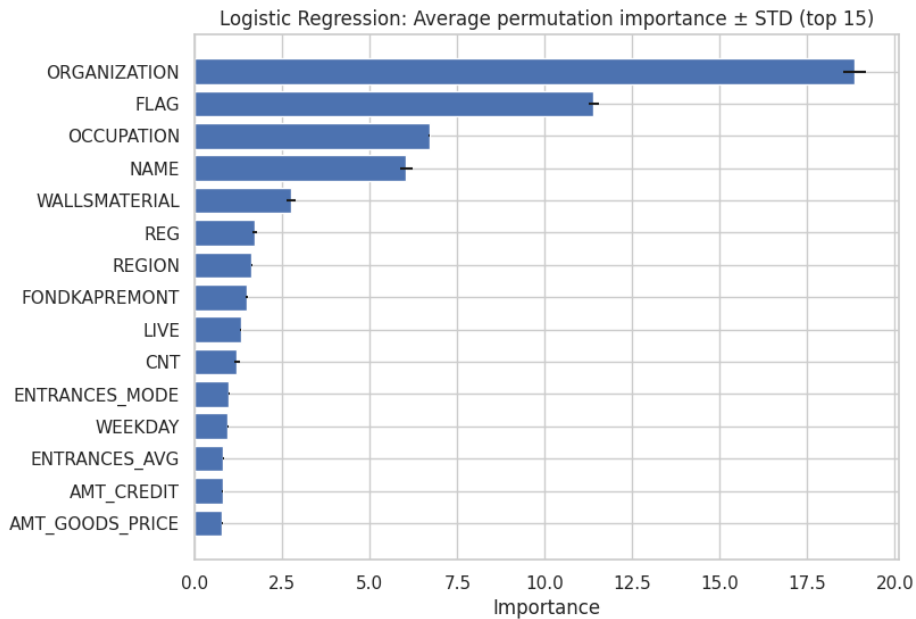


Figure 5.2: Average permutation importance for logistic regression with a spread over launches.

For the decision tree, feature importance was less stable, with correlations around 0.8 and a more noticeable change in feature ranking across runs. This was expected, since the tree depends more strongly on the specific sample and on the split thresholds chosen at the upper levels. Still, external scoring indicators consistently remained among the most important features.

LightGBM showed a more complex but still fairly consistent pattern. Correlations between runs ranged from about 0.75 to 0.9, meaning that the ranking changed more than in logistic regression, but the main factors stayed the same (Figure 5.3). Both gain importance and SHAP values highlighted similar leading features, including EXT_SOURCE_*, age, and debt burden indicators. The overlap of the top 10 features across runs was about 80%, which indicates a fairly good level of consistency for a more complex model.

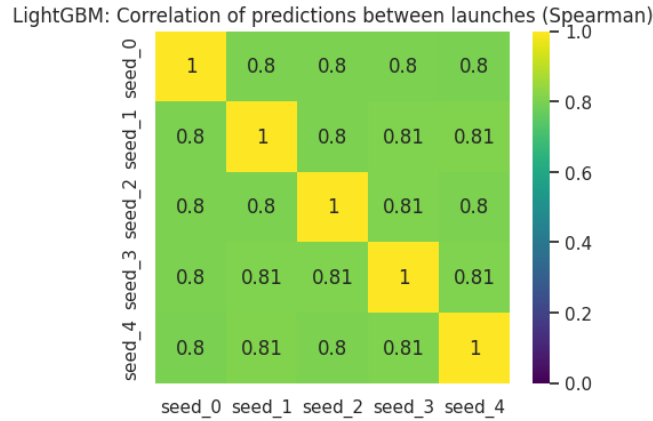


Figure 5.3: Heatmap of correlation of importance of logistic regression between launches.

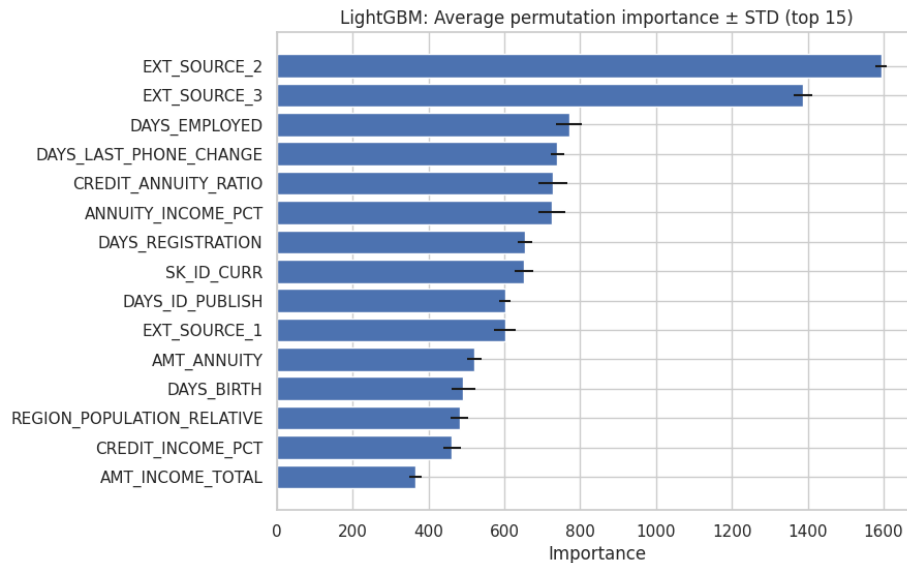


Figure 5.4: Average permutation importance for logistic regression with a spread over launches.

The results lead to an important conclusion: despite differences in stability, all models share a common core of key features. These include external scoring indicators, age, loan parameters, and debt burden variables. Thus, even if the ranking of individual features changes slightly, the overall interpretation remains stable (Figures 5.4, 5.5).

Overall, logistic regression showed the highest stability of feature importance, the decision tree was more sensitive to sample changes, and LightGBM showed moderate variation while preserv-

ing the same main factors. This suggests that the interpretation results are generally reproducible and reliable.

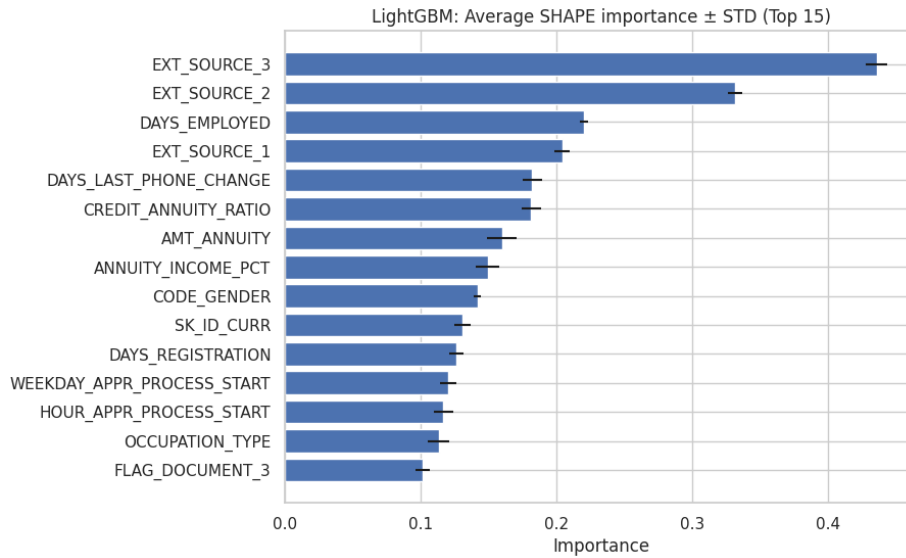


Figure 5.5: Average permutation importance for logistic regression with a spread over launches.

5.5 Analysis of the stability of model predictions

The final stage of the stability assessment focused on prediction consistency. To measure it, Spearman rank correlation was calculated between probabilistic predictions from different runs. High correlation meant that the model ranked borrowers by risk in a similar way despite changes in the data split and `random_state`. Logistic regression showed the most stable predictions, with correlations above 0.97. This indicates that the model produced highly consistent probabilistic estimates across repeated runs, which is in line with its previously observed stability in both metrics and feature importance (Figures 5.1).

For the decision tree, prediction stability was lower, with noticeably weaker correlations between runs than in logistic regression. This indicates greater sensitivity to changes in the data, which is expected because even small shifts in the sample can alter split thresholds and tree structure. As a result, the decision tree was less reproducible not only in terms of metrics and feature importance, but also in its predictions. **LightGBM** also showed high prediction consistency, with correlations of about 0.96. This means that the boosting model ranked clients by risk in a stable way, even if some probability estimates changed slightly across runs (Figure 5.3).

Overall, logistic regression produced the most reproducible predictions, **LightGBM** remained highly stable with only a slight decrease in consistency, and the decision tree was the most sensitive to data changes.

6 LLM-assisted interpretation of credit scoring results

6.1 Using a language model to explain credit scoring results

The final additional stage of the work was the construction of an LLM-assisted explanation layer. Unlike logistic regression, decision tree, `LightGBM`, and `MLP` models, the language model was not used to predict the probability of default. Its task was to transform the already obtained model result into a clear textual explanation.

The general logic of this block was based on the output of the scoring model and local interpretation methods. First, the trained model estimated the probability of default for a selected application. Then `SHAP` values were used to identify the features that increased or decreased the predicted risk. After that, the predicted probability, risk category, and the most important local factors were passed to the language model.

In the experiment, `LightGBM` was used as the base scoring model. For the analyzed application, the predicted probability of default was equal to 0.0538, which corresponded to a moderate risk category. The local explanation showed that some factors reduced the estimated risk, while others increased it. Among the risk-reducing factors were `AMT_GOODS_PRICE`, `NAME_EDUCATION_TYPE`, `CREDIT_ANNUITY_RATIO`, `EXT_SOURCE_2`, and `FLAG_OWN_CAR`. At the same time, `DAYS_EMPLOYED`, `EXT_SOURCE_3`, `ORGANIZATION_TYPE`, `CODE_GENDER`, and the missing value of `EXT_SOURCE_1` shifted the prediction toward a higher risk level.

The language model used this structured information to generate a short explanation in business language. The prompt was designed so that the model relied only on the provided facts, did not invent additional client characteristics, did not make causal claims and did not make the final credit decision. This limitation is important because the LLM should not replace the scoring model or the analyst.

7 Conclusion

The thesis considered the task of constructing an interpretable credit scoring model based on anonymized dataset close to real data. The focus was not only on the quality of the forecast, but also on the transparency of the model, as well as the sustainability of the results obtained during repeated launches. In the course of the work, data was prepared, omissions and abnormal values were processed, derived features were formed, variables were divided into numeric and categorical, and a single preprocessing pipeline was built. Additionally, different approaches to coding categorical features were compared and the most informative factors were selected.

Model building has shown that LightGBM provides the best forecast quality. Logistic regression showed slightly more modest, but at the same time quite high results and retained the best interpretability. The decision tree was inferior in terms of quality metrics, but it turned out to be useful in terms of visual representation of the logic of the scoring solution.

The interpretation of the results showed that external scoring indicators, signs of debt burden, loan parameters and part of the demographic characteristics of the client have the greatest impact on the probability of default. These conclusions were consistently repeated in different models, which increases the credibility of the results obtained.

The stability check showed that logistic regression behaves most smoothly and reproducibly, and LightGBM combines high quality with acceptable stability. As a result, we can conclude that the interpreted approach to the problem of credit scoring allows us to obtain not only a good level of forecast, but also a meaningfully explicable result, which is especially important for practical application in the financial sector.

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